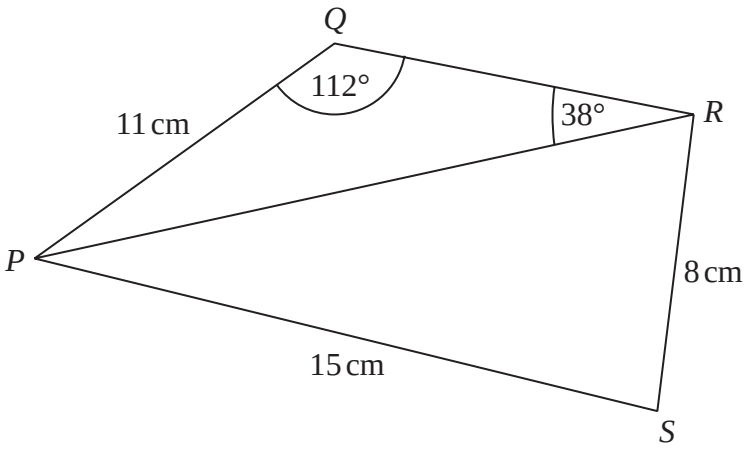
13 *PQR* and *PRS* are triangles.



Calculate the size of angle *PSR*.
Give your answer correct to 1 decimal place.

In triangle PQR, use the sine rule:

$$\frac{PR}{\sin 112°} = \frac{11}{\sin 38°}$$

$PR = 11 \times \sin 112° \div \sin 38° = 16.565958\ldots$ cm

2 marks total
Match each side with its opposite angle; retain full precision.

Let θ = angle PSR. In triangle PSR:

$PR^2 = 8^2 + 15^2 - 2 \times 8 \times 15 \times \cos\theta$

$\cos\theta = (8^2 + 15^2 - PR^2) \div (2 \times 8 \times 15)$

$\theta = \cos^{-1}(0.0607\ldots) = 86.5°$ (1 d.p.)

3 marks total
Use cosine rule with all three sides; round the final angle.

86.5 °

(Total for Question 13 is 5 marks)

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